

TABLE-II
LED vs Traditional Lamps

Lamp	CCT(K)	Response time	Light-out theory	Light direction	Life	Reliability	Vibration resistance	Disadvantages
LED	2600-10,000	<1 sec.	Semiconductor	Specific direction	50,000 hours	High	High	High price
CFL	4000-8000	<60 sec.	Hg vapour	Full direction	6000 hours	Middle	Low	Hg pollution, broken
Halogen	2500-3000	<150 sec.	Heating	Full direction	1000 hours	Low	Low	Power exhaust, short life, fragile

into light and only some of it is dissipated as heat.

Small form-factor. LEDs are very small. Typically, high-brightness LED chips measure $0.3 \times 0.3 \text{ mm}^2$, while high-power devices can be $1 \times 1 \text{ mm}^2$ or larger. There are many examples where the availability of small, high-brightness devices have enabled significant market advancement. One obvious example is mobile phone handsets, where blue, green and white LEDs are now used in most models to backlight keypads and liquid-crystal display (LCD) screens.

Instantaneous switch-on. LEDs switch-on rapidly, even when cold, and this is a particular advantage for certain applications such as vehicle brake lights. These light up very quickly and achieve full brightness within a few microseconds.

Colour. LEDs are available in a broad range of brilliant, saturated colours. White devices are also available.

Also, LED lights are shock-resistant, and a solid package of the LEDs can be designed to focus their light. The fact that LEDs are non-toxic (do not contain mercury) also favours their application.

On the downside, in many applications, LEDs prove to be expensive compared to other light sources. However, LED manufacturers are working towards reducing their cost.

LED applications

LED applications can be categorised into visual signal applications (where the light moves from the bulb to the human eyes directly), illuminations (where LED light is reflected from an object to give visual response) and technological (which don't require

contact with the human visual system as in the case of light photosynthesis in plants).

LED lights are being used in many areas, such as LCD screens, mobile phones, iPods and safety and security hardware. LED technology is vast and growing by leaps and bounds. The newest trend in the electronics market is the green influence that has not left the LED world bereft. The most recent launches in the LED market are the light bulbs that can help you save up to 90 per cent of electricity costs and reduce the heat output. Moreover, these bulbs are claimed to last up to ten times longer than a standard light bulb or compact fluorescent light. Some of the most advanced LED-emitting products in the electronics world are LED jellyfish mood lamps, Sony Walkmans and hydroponic LED lights.

LED jellyfish mood lamps. These are unique desktop tanks that trick your eyes into believing that three jellyfish are swimming, with the aid

of six bright colourful LEDs that are placed at the top of the tank.

Sony's Walkman. Sony has launched a Walkman called NWZ-X1000 that features an organic light-emitting diode (OLED) touch-screen.

Hydroponic LED lights. Growing plants indoor with a hydroponic system is becoming popular. Plants take the light and turn it into energy through the process of photosynthesis. This is not just an option but a necessity and will ensure that you have plants that thrive and give you the results that you are looking for. Luckily, there are ways to fool plants into thinking that they have been given all the light they need, and for this you can use LED lights.

LED lighting is also used for security reasons and as traffic lights. Since these lights consume very little power, they are ideal for long running operations or where the lights are required to remain on for an indefinite time period.

Basically, LEDs are available in three categories: LEDs for cars, LED flashlights, and LED bulbs or strips for home. LED tubes with different patterns are also becoming popular. LED rope light for household use is yet another wonderful light that you can use for home décor. These lights look pleasing to the eye and can be used to outline the boundary of an object.

LED car kits are admired by a lot of car enthusiasts all over the world. These LED lights create an under-car halo. LED car kits require 12V power and come with 2.1-million colour output. These can display one colour at a time and the pattern can be chosen according to the terrain or situation. The lights are waterproof and can last for years to come.



LED jellyfish mood lamp